Lab Environment Setup Report

NexaSecure Ethical Hacking Internship
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1. Introduction:

This report describes the setup of a complete cybersecurity lab environment for ethical hacking practice using virtualization.

2. Tools Used

- Kali Linux
- VirtualBox / VMware
- Metasploitable 2
- Linux Terminal Utilities

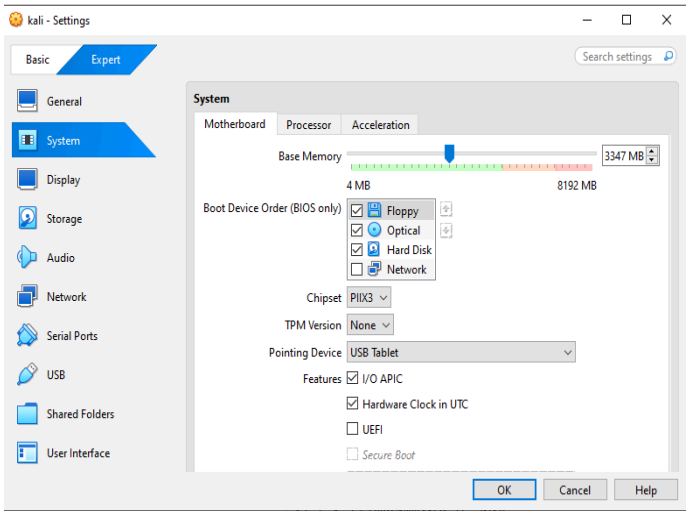
3. System Configuration

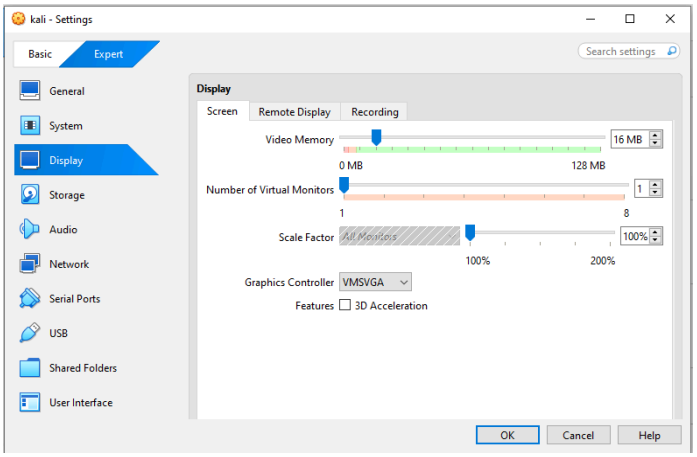
Host OS: Windows 10

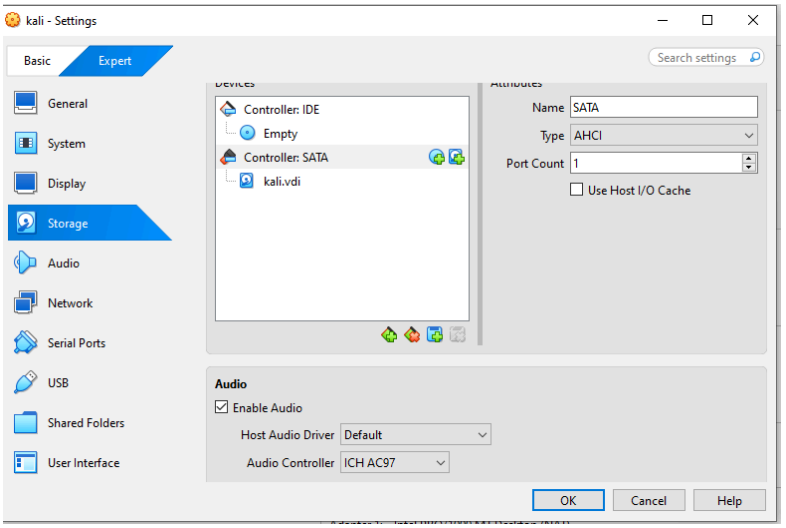
RAM Allocated to VM: 3347MB

Disk Size: 20GB

Network Mode: Bridged

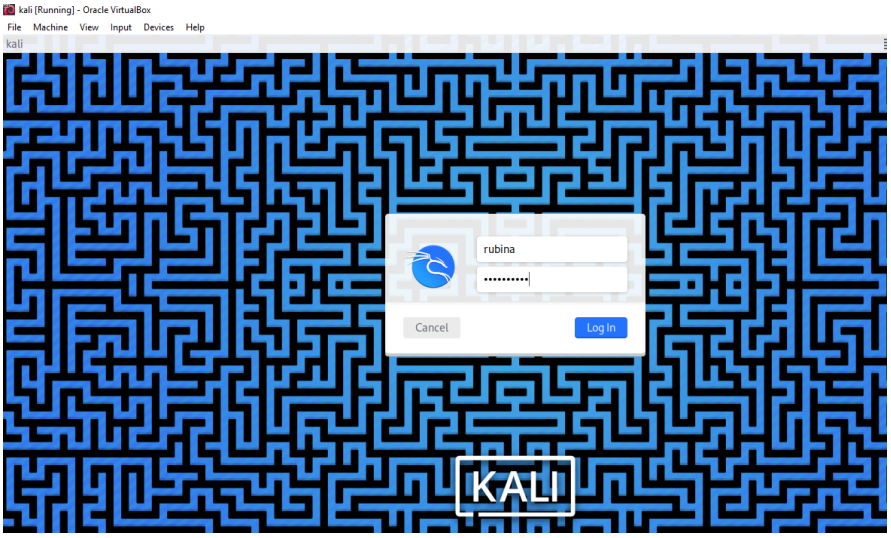


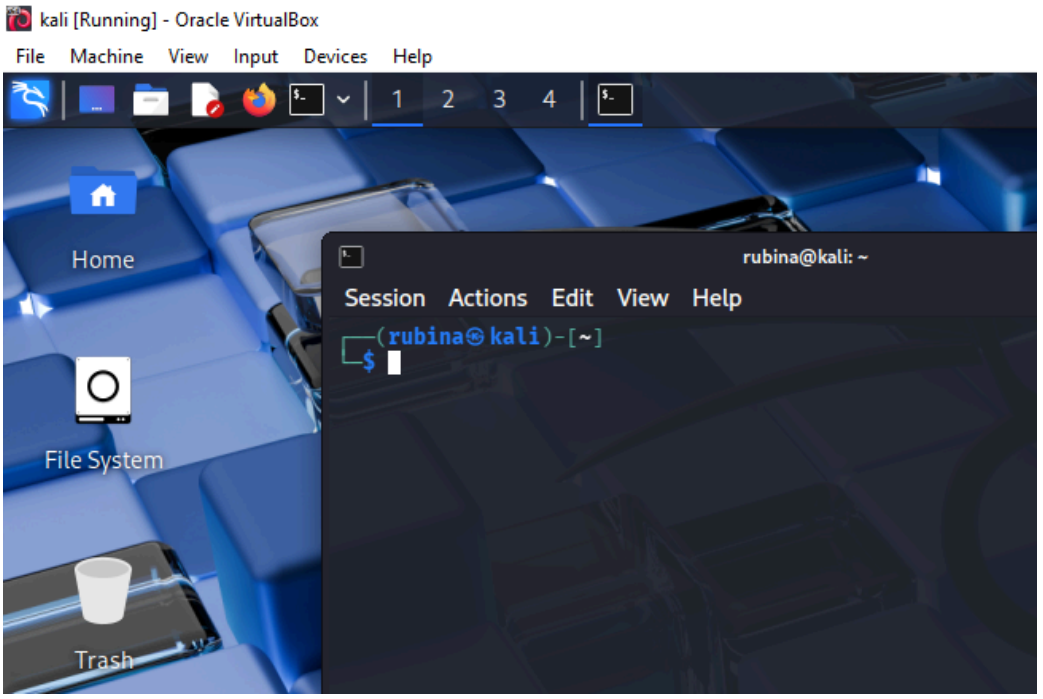


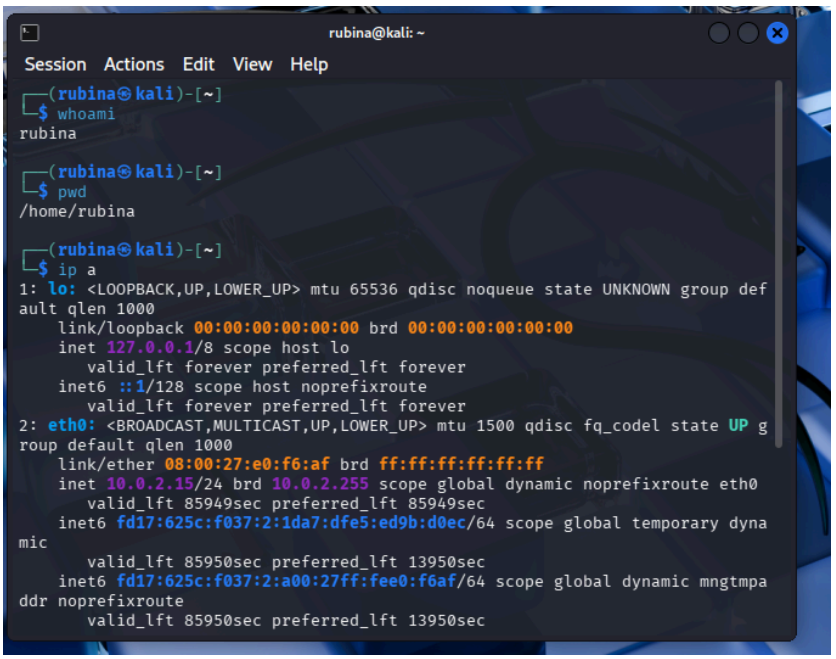


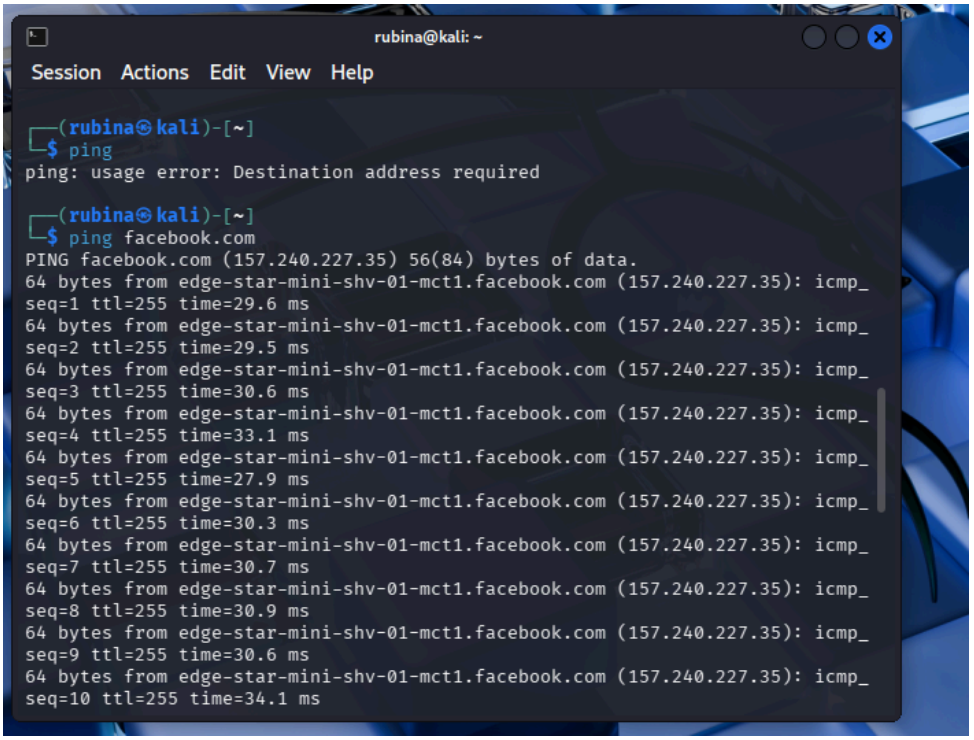
4. Kali Linux Installation

Kali Linux was installed inside a virtual machine environment. The installation was completed successfully and system was verified using terminal commands.


kali [Running] - Oracle VirtualBox
File Machine View Input Devices Help
kali
rubina
Cancel
Log In
KALI


kali [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4
Home
File System
Trash
rubina@kali: ~
Session Actions Edit View Help
(rubina㉿kali)-[~]
$

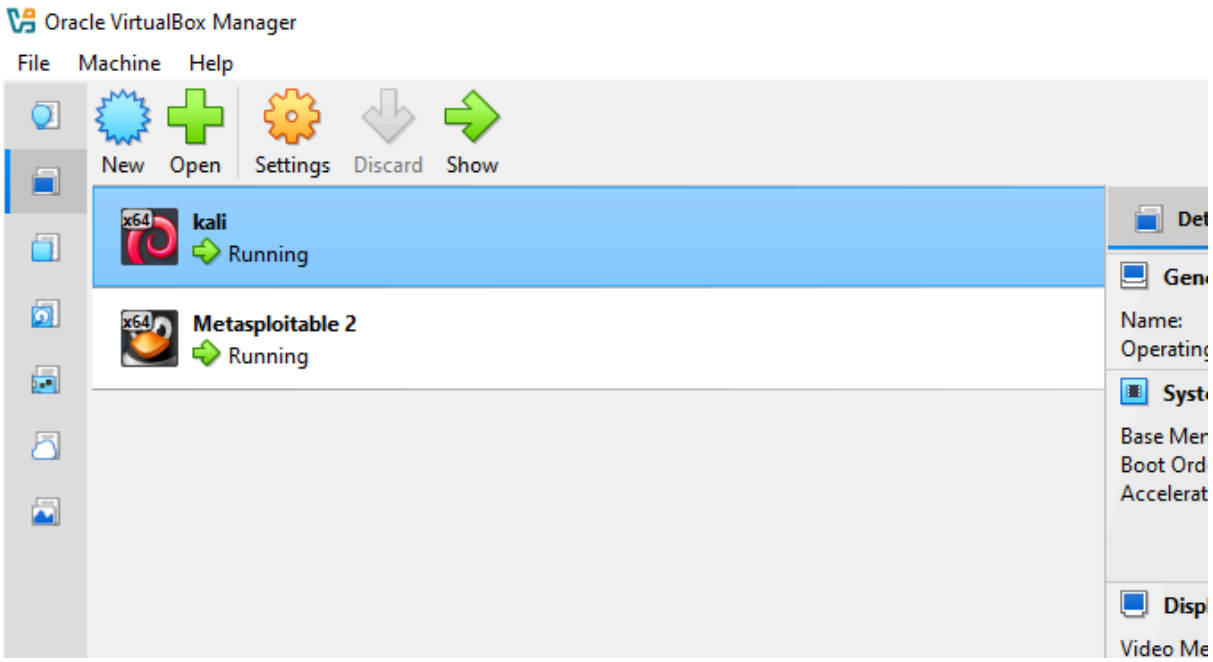


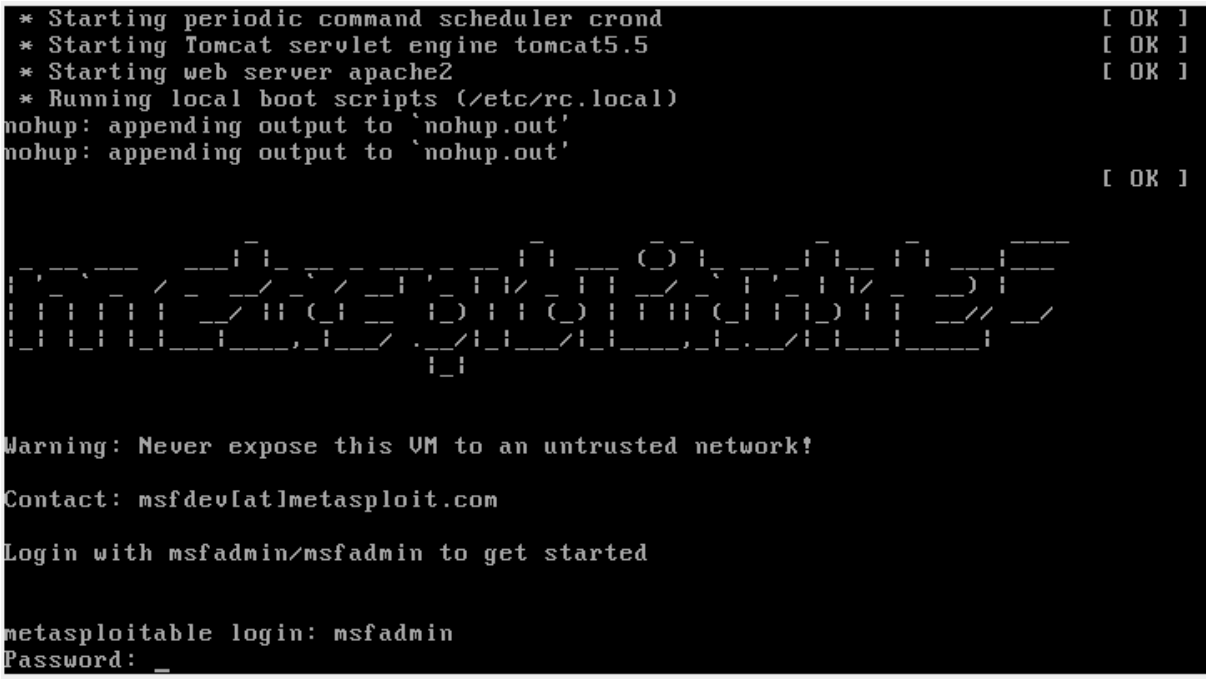


5. Metasploitable 2 Setup:

I have installed Metasploitable 2 in Oracle VirtualBox, configured networking between Kali Linux and Metasploitable 2, and verified connectivity for penetration testing practice.

Downloaded Metasploitable 2 VM file.
Imported into VirtualBox.
Started VM.
Default credentials used:
Admin: msfadmin
Password: msfadmin





Check IP Address of Metasploitable

Command:
Ifconfig

Output:
192.168.1.3

```
To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:7f:d6:56
          inet addr:192.168.1.3  Bcast:192.168.1.255  Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:fe7f:d656/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:22 errors:0 dropped:0 overruns:0 frame:0
          TX packets:51 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:3338 (3.2 KB)  TX bytes:5622 (5.4 KB)
          Base address:0xd020 Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:101 errors:0 dropped:0 overruns:0 frame:0
          TX packets:101 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:23573 (23.0 KB)  TX bytes:23573 (23.0 KB)

msfadmin@metasploitable:~$ _
```

Test Connectivity from Kali Linux

Command used in Kali:

Ping 192.168.1.3

Result:

- Successful replies received.
- No packet loss.
- TTL=64
- Time < 1ms

This confirms that Kali Linux and Metasploitable 2 are properly connected.

rubina@kali: ~

Session Actions Edit View Help

```
┌──(rubina㉿kali)-[~]
└─$ ping 192.168.1.3
PING 192.168.1.3 (192.168.1.3) 56(84) bytes of data.
64 bytes from 192.168.1.3: icmp_seq=1 ttl=64 time=0.556 ms
64 bytes from 192.168.1.3: icmp_seq=2 ttl=64 time=0.367 ms
64 bytes from 192.168.1.3: icmp_seq=3 ttl=64 time=0.290 ms
64 bytes from 192.168.1.3: icmp_seq=4 ttl=64 time=0.303 ms
64 bytes from 192.168.1.3: icmp_seq=5 ttl=64 time=0.511 ms
64 bytes from 192.168.1.3: icmp_seq=6 ttl=64 time=0.478 ms
64 bytes from 192.168.1.3: icmp_seq=7 ttl=64 time=1.07 ms
64 bytes from 192.168.1.3: icmp_seq=8 ttl=64 time=0.966 ms
64 bytes from 192.168.1.3: icmp_seq=9 ttl=64 time=0.948 ms
64 bytes from 192.168.1.3: icmp_seq=10 ttl=64 time=0.887 ms
64 bytes from 192.168.1.3: icmp_seq=11 ttl=64 time=0.891 ms
64 bytes from 192.168.1.3: icmp_seq=12 ttl=64 time=0.835 ms
64 bytes from 192.168.1.3: icmp_seq=13 ttl=64 time=0.352 ms
64 bytes from 192.168.1.3: icmp_seq=14 ttl=64 time=0.909 ms
64 bytes from 192.168.1.3: icmp_seq=15 ttl=64 time=0.548 ms
64 bytes from 192.168.1.3: icmp_seq=16 ttl=64 time=0.368 ms
64 bytes from 192.168.1.3: icmp_seq=17 ttl=64 time=0.331 ms
64 bytes from 192.168.1.3: icmp_seq=18 ttl=64 time=0.485 ms
64 bytes from 192.168.1.3: icmp_seq=19 ttl=64 time=0.716 ms
64 bytes from 192.168.1.3: icmp_seq=20 ttl=64 time=0.980 ms
64 bytes from 192.168.1.3: icmp_seq=21 ttl=64 time=0.662 ms
64 bytes from 192.168.1.3: icmp_seq=22 ttl=64 time=0.935 ms
64 bytes from 192.168.1.3: icmp_seq=23 ttl=64 time=0.455 ms
64 bytes from 192.168.1.3: icmp_seq=24 ttl=64 time=0.728 ms
64 bytes from 192.168.1.3: icmp_seq=25 ttl=64 time=0.698 ms
```